Grouped Diagonals and Arithmetic Bounds in Creative Telescoping

Constructive proofs addressing Chen–Kauers Problems 12 and 4

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Abstract

We give two constructive statements associated with the problem list of Chen and Kauers. First, multivariate binomial sums over a field of characteristic zero are exactly grouped diagonals of rational power series. The proof converts the monomial-section representation of Bostan, Lairez, and Salvy into disjoint blocks of repeated indices by a classical coefficient-splitting identity. A representation in d variables with s free indices can be converted using at most ds variables when $d \geq 1$. Second, for a hyperexponential function with rational logarithmic derivatives, we give explicit integer coefficient bounds for a polynomial telescoper and its rational certificate. With logarithmic-derivative bidegrees bounded by (t, s), a possibly nonminimal telescoper can be chosen with coefficient bit length $O(ts^3(h + \log(ts + 2)))$, where h measures the input coefficient norms and $t, s \geq 1$. All exceptional cases in the construction are treated. The proofs use established algebraic techniques; the manuscript does not certify historical priority or a new foundational method. Exact supplementary checks are embedded in the single TeX source.

1 Questions, results, and attribution

This note uses the original arXiv numbering in Chen–Kauers [1]. Problem 12 asks for a characterization of multivariate binomial sums by coefficients of rational power series with each free index repeated in a separate block. Problem 4 asks for bounds on the bit size of hyperexponential telescopers. Their journal numbers are, respectively, 6.3 and 3.2.

Theorem 2.1 proves the former equivalence for sequences on $\mathbb{N}^s$, the index domain implicit in ordinary power series. Theorem 3.1 gives explicit bit bounds for every hyperexponential input whose logarithmic derivatives belong to $\mathbb{Q}(x, y)$, under a stated finite arithmetic representation. The latter theorem asserts existence at bounded order and degree, without requiring the order to be minimal.

Dependence on earlier work. The substantial rational monomial-section construction in the first argument is due to Bostan–Lairez–Salvy [2]. Its remaining conversion is proved here using the classical Hankel-array identity recorded, for example, by Barry [3, p. 5]. The second argument uses polynomial ansatz, dimension counting, and Hadamard bounds. These techniques have established precedents in creative telescoping [4, 5]. The resulting statements should be assessed as explicit constructive derivations, not evidence that their underlying tools originated here.

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Priority and validation. A search of the primary sources considered in preparing this note did not locate an explicit identical full result. That search cannot establish the absence of prior work, and no claim of first resolution is made. The two proofs were checked independently during preparation; the additional computations are exact rational or integer checks, not formal proof-assistant verification or external peer review.

The distinction between algebraic complexity and coefficient bit size matters for the second question. Van der Hoeven's general reduction framework [6, Section 8.1 and Theorem 9.8] measures field operations and coefficient counts. Its stated polynomial complexity does not by itself provide the explicit integer coefficient bounds proved below. This observation does not imply that no other source contains such bounds.

2 Grouped diagonals and multivariate binomial sums

Chen and Kauers ask whether the multivariate binomial sums of Bostan, Lairez, and Salvy admit rational representations in which each free index is repeated in its own block of variables [1, Problem 12]. The journal version labels this question Problem 6.3. We give an affirmative proof by converting a nonnegative monomial section into a grouped diagonal. The input representation is due to Bostan, Lairez, and Salvy; the conversion uses an elementary divided-difference operation on coefficient indices.

Throughout this section, K is a field of characteristic zero and $\mathbb{N} = \{0, 1, 2, \ldots\}$. A *rational power series* is an element of $K(\boldsymbol{z}) \cap K[[\boldsymbol{z}]]$ represented as P/Q with $P, Q \in K[\boldsymbol{z}]$ and $Q(\mathbf{0}) \neq 0$. We use the binomial-sum class of [2, Definition 1.1]: it is generated by Kronecker deltas, nonzero geometric sequences, and binomial coefficients, under addition, multiplication, affine integer changes of indices, and directed indefinite summation. A sequence on $\mathbb{N}^s$ belongs to this class when its zero extension to $\mathbb{Z}^s$ does. This convention is legitimate because the indicator $H(n) = \mathbf{1}_{n \geq 0}$ is itself a binomial sum.

The nonnegative index set is the one implicit in the rational-power-series formulation of the question. A two-sided sequence on $\mathbb{Z}^s$ has additional negative-index data, which a single ordinary power series does not encode. We do not reinterpret those data as coefficients with negative exponents.

Theorem 2.1 (Grouped-diagonal characterization)**.** *Let $s \geq 1$ and let $b\colon \mathbb{N}^s \longrightarrow K$. The following conditions are equivalent.*

1. *The sequence b is a binomial sum.*
2. *There exist positive integers $r_1, \ldots, r_s$ and a rational power series U in variables $x_{j,\ell}$, $1 \leq j \leq s$, $1 \leq \ell \leq r_j$, such that, for every $\boldsymbol{n} \in \mathbb{N}^s$,*

$$b(\boldsymbol{n}) = \left[\prod_{j=1}^{s} \prod_{\ell=1}^{r_j} x_{j,\ell}^{n_j} \right] U. \tag{1}$$

Both implications are constructive.

2.1 The existing monomial-section representation

We first state precisely the result from which the conversion starts. It is the intermediate construction in the proof of [2, Proposition 3.10 and Section 3.1].

Proposition 2.2 (Bostan–Lairez–Salvy monomial-section input)**.** *For every binomial sum $b\colon \mathbb{N}^s \to K$, one can construct a rational power series*

$$T(\boldsymbol{z}) = \sum_{\boldsymbol{m} \in \mathbb{N}^d} c_{\boldsymbol{m}} \boldsymbol{z}^{\boldsymbol{m}}$$

and a matrix $A \in \mathbb{N}^{d\times s}$ *such that*

$$b(\boldsymbol{n}) = c_{A\boldsymbol{n}} \qquad (\boldsymbol{n} \in \mathbb{N}^s). \tag{2}$$

For clarity, the nonnegativity and origin regularity in this statement are substantial parts of the existing construction. The initial representation is a constant term of a rational function in an ordered Laurent-series field. The nonnegative integral shear of [2, Lemma 3.11] turns its denominator into $C\boldsymbol{z}^{\boldsymbol{\mu}}(1+G)$, with $C \neq 0$, $\boldsymbol{\mu} \in \mathbb{N}^d$, and $G(\mathbf{0}) = 0$. The shear is injective and respects the chosen Laurent-series expansion; the target monomials therefore become $\boldsymbol{z}^{A\boldsymbol{n}}$ with $A \geq 0$. At this stage the representation is

$$b(\boldsymbol{n}) = [\boldsymbol{z}^{A\boldsymbol{n}}]\boldsymbol{z}^{-\boldsymbol{\mu}}S(\boldsymbol{z}), \qquad S \text{ rational and regular at } \mathbf{0}.$$

If $\mu_i > 0$, replace S by $(S - S|_{z_i=0})/z_i$ and reduce μ_i by one. This preserves every target coefficient: the discarded term has exponent $-\mu_i < 0$ in z_i, whereas $(A\boldsymbol{n})_i \geq 0$. Each replacement is rational and regular, and exactly $\sum_i \mu_i$ such replacements suffice. This gives (2). This explanation also shows why the one-index argument in Section 3.1 applies to the several target monomials in Proposition 3.10.

The columns of A can overlap in their nonzero rows. Consequently, (2) is not already the repeated-block representation (1). The following operation removes that overlap.

2.2 Splitting coefficient indices

The identity below is the familiar generating function of a Hankel array; it appears explicitly, for example, in [3, p. 5]. We include its proof with auxiliary variables and origin regularity, which are needed for the repeated construction.

Lemma 2.3 (Binary splitting)**.** *Let* $F(z, \boldsymbol{u})$ *be rational and regular at the origin, with*

$$F(z, \boldsymbol{u}) = \sum_{n\geq 0} f_n(\boldsymbol{u})z^n.$$

Then

$$\mathcal{S}_z F(x, y, \boldsymbol{u}) = \frac{xF(x, \boldsymbol{u}) - yF(y, \boldsymbol{u})}{x - y} = \sum_{p,q\geq 0} f_{p+q}(\boldsymbol{u})x^p y^q \tag{3}$$

is also rational and regular at the origin.

Proof. The coefficient identity follows from

$$\frac{x^{n+1} - y^{n+1}}{x - y} = \sum_{p+q=n} x^p y^q.$$

To check regularity rather than just rationality, write $F = P/Q$ with $Q(0, \mathbf{0}) \neq 0$. Over the denominator $Q(x, \boldsymbol{u})Q(y, \boldsymbol{u})$ the numerator before division by $x - y$ is

$$xP(x, \boldsymbol{u})Q(y, \boldsymbol{u}) - yP(y, \boldsymbol{u})Q(x, \boldsymbol{u}).$$

This polynomial vanishes at $x = y$, hence is divisible by $x - y$. After cancellation the denominator is a unit at the origin. Thus the apparent pole along $x = y$ is removable by polynomial division. The identity is coefficientwise and requires no interchange of infinite sums with infinitely many contributors to a coefficient. □

Repeated splitting replaces a coefficient index m by a sum of any prescribed positive number q of independent indices. With $q = 1$ nothing is changed, and with $q = 0$ the appropriate operation is specialization of the original variable to zero.

Lemma 2.4 (Monomial sections as grouped diagonals)**.** *Let $T = \sum_{\boldsymbol{m}\in\mathbb{N}^d} c_{\boldsymbol{m}}\boldsymbol{z}^{\boldsymbol{m}}$ be a rational power series, and let $A \in \mathbb{N}^{d\times s}$. The sequence $c_{A\boldsymbol{n}}$ has a representation of the form (1). Such a representation can be constructed using*

$$\sum_{i=1}^{d}\sum_{j=1}^{s} A_{ij} + \#\{j : A_{ij} = 0 \text{ for every } i\}$$

variables.

Proof. For each row i, put $q_i = \sum_j A_{ij}$. If $q_i = 0$, specialize $z_i = 0$. Otherwise use Lemma 2.3 to replace z_i by q_i independent variables, labelled

$$x_{i,j,\ell}, \qquad 1 \le j \le s, \quad 1 \le \ell \le A_{ij}.$$

Carry this out for all rows, obtaining a regular rational series V. For arbitrary nonnegative exponents $k_{i,j,\ell}$ its coefficients are

$$\left[\prod_{i,j,\ell} x_{i,j,\ell}^{k_{i,j,\ell}}\right] V = c_{\boldsymbol{m}}, \qquad m_i = \sum_{j,\ell} k_{i,j,\ell}. \tag{4}$$

In particular, requiring $k_{i,j,\ell} = n_j$ gives $m_i = \sum_j A_{ij}n_j$. All variables with middle label j form the jth block. If column j is zero, multiply V by $(1 - v_j)^{-1}$ in a fresh variable and use $\{v_j\}$ as that block. Its coefficient is one at every nonnegative index, so the desired sequence is unchanged. All blocks are now nonempty. Reordering the variables gives (1), with the asserted variable count. The constant case $d = 0$ is handled by $U = T\prod_{j=1}^{s}(1 - v_j)^{-1}$. □

2.3 Proof of the characterization

Proof of Theorem 2.1. For the forward implication, apply Proposition 2.2 and then Lemma 2.4.

For the converse we give an explicit finite-sum argument. By scaling the denominator and separating the monomials of the numerator, it suffices to consider

$$R(\boldsymbol{z}) = \frac{\boldsymbol{z}^{\boldsymbol{\beta}}}{1 - \sum_{\ell=1}^{e} a_\ell \boldsymbol{z}^{\boldsymbol{\alpha}_\ell}}, \tag{5}$$

where $\boldsymbol{\beta} \in \mathbb{N}^r$, $\boldsymbol{\alpha}_\ell \in \mathbb{N}^r \setminus \{\mathbf{0}\}$, and $a_\ell \in K \setminus \{0\}$. Its coefficient at $\boldsymbol{z}^{\boldsymbol{m}}$ equals

$$\sum_{\substack{\boldsymbol{k}\in\mathbb{N}^e \\ \boldsymbol{\beta}+\sum_\ell k_\ell\boldsymbol{\alpha}_\ell=\boldsymbol{m}}} \binom{k_1 + \cdots + k_e}{k_1, \ldots, k_e} \prod_{\ell=1}^{e} a_\ell^{k_\ell}. \tag{6}$$

Every contributing tuple satisfies $\sum_\ell k_\ell \le |\boldsymbol{m}|_1$, because each $|\boldsymbol{\alpha}_\ell|_1 \ge 1$. Thus (6) can be written as nested finite sums with bounds $0 \le k_\ell \le |\boldsymbol{m}|_1$, inserting one Kronecker delta for each coordinate of the displayed equality. These bounds are affine in $\boldsymbol{m}$. The multinomial factor is a product of binomial coefficients, and the remaining factors are permitted geometric sequences and deltas. The defining closure rules therefore make the entire coefficient array a binomial sum on $\mathbb{N}^r$. Multiplication by $\prod_i H(m_i)$ supplies its zero extension. When $e = 0$, the coefficient array is simply a product of deltas, so the same conclusion holds. Finally, the repeated-block map from $\boldsymbol{n}$ to $\boldsymbol{m}$ is an affine integer substitution, under which binomial sums are closed. This proves the converse and also its effectivity. □

2.4 Example and a smaller variable bound

Take $T(z,w) = (1-z-w)^{-1}$ and

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

The corresponding monomial section is $b(m,n) = \binom{2m+n}{m}$. Splitting the z index gives

$$U(a,b,c) = \frac{1-c}{(1-a-c)(1-b-c)}, \qquad [a^m c^m b^n]U = \binom{2m+n}{m}. \tag{7}$$

Indeed, $[a^m b^n]U = (1-c)^{-m-n-1}$. Thus the shared index $m+n$ in the original z coordinate becomes two independent variables, subsequently assigned to different blocks.

The copies-only construction above is sufficient for the theorem. For large entries of A, ordinary rational multisections reduce its number of variables.

Proposition 2.5 (A dimension bound)**.** *If $d \geq 1$, the representation in Lemma 2.4 can be constructed with at most ds variables, independently of the sizes of the entries of A.*

Proof. Split row i into one variable x_{ij} for each nonzero entry A_{ij}, rather than into $\sum_j A_{ij}$ variables. The resulting coefficient array is $c_{(\sum_j k_{1j},\ldots,\sum_j k_{dj})}$. In coordinate x_{ij} take the A_{ij}-multisection: retain the coefficients whose exponent is divisible by A_{ij} and divide that exponent by A_{ij}.

To justify this operation over K, let $F(x,\boldsymbol{u}) = P/Q$ be regular at the origin and let $a \geq 1$. In a finite splitting field for the ath roots of unity form

$$W(x,\boldsymbol{u}) = \frac{1}{a}\sum_{\zeta^a=1} F(\zeta x,\boldsymbol{u}).$$

Its series contains exactly the terms with x exponent divisible by a. A denominator for W is $D = \prod_{\zeta^a=1} Q(\zeta x,\boldsymbol{u})$, whose constant term is nonzero. Both D and the polynomial numerator DW are invariant under $x \mapsto \zeta x$ and under the Galois group of the splitting field. Hence both belong to $K[x^a,\boldsymbol{u}]$. Replacing x^a by a new variable gives a rational power series over K with coefficient $[x^{an}]F$ at exponent n.

Apply these multisections independently. The coefficient array becomes

$$c_{(\sum_j A_{1j}k_{1j},\ldots,\sum_j A_{dj}k_{dj})}.$$

Its grouped diagonal is the required sequence. Add one geometric dummy variable for each zero column, as before. If there are q zero columns, the variable count is at most $d(s-q)+q \leq ds$. □

Remark 2.6 (Scope of the contribution). The proof uses the established Bostan–Lairez–Salvy representation and classical divided differences and multisections. Its additional step is the explicit conversion from overlapping nonnegative monomial sections to disjoint repeated-index blocks. This supplies a proof of the stated characterization; it does not, by itself, establish publication priority or a historically new foundational principle.

3 An explicit arithmetic bound for hyperexponential telescopers

The original arXiv Problem 4 of Chen and Kauers [1], numbered Problem 3.2 in the journal version, asks for bounds on the bit size of telescopers for hyperexponential terms. We give an explicit bound in the rational logarithmic-derivative input model. The proof is self-contained, but its methodology is classical: polynomial ansatz and dimension counting, as in the order–degree analysis of Chen and Kauers [4], followed by integer-kernel bounds using determinants.

Arithmetic size analysis for hypergeometric telescopers was already undertaken by Kauers and Yen [5]. No claim of a historically new method or established priority is made here.

Let $\mathcal{F}$ be a differential field extension of $\mathbb{Q}(x,y)$ with commuting derivations ∂_x, ∂_y, and let $0 \neq f \in \mathcal{F}$. Suppose we are given $A, B, D \in \mathbb{Z}[x,y]$, $D \neq 0$, such that

$$\frac{\partial_x f}{f} = \frac{A}{D}, \qquad \frac{\partial_y f}{f} = \frac{B}{D}. \tag{8}$$

Every pair of logarithmic derivatives in $\mathbb{Q}(x,y)$ admits this representation by clearing polynomial and integer denominators. For $F \in \mathbb{Z}[x,y]$, write $\|F\|_1$ for the sum of the absolute values of its coefficients. The coefficient magnitude bit length of a nonzero integer polynomial operator is $1 + \lfloor \log_2 \max |c| \rfloor$, where c runs over its integer coefficients; sign bits are counted separately.

Theorem 3.1. *Let $t, s \geq 0$ be integer upper bounds on the respective x- and y-degrees of A, B, D. Let r, d be nonnegative integers satisfying*

$$r > s, \qquad (r-s)(d+1) > t\big((2r-1)s + r\big). \tag{9}$$

Put

$$\begin{aligned} a &= d + (r-1)t, & b &= (r-1)s, \\ n_P &= (r+1)(d+1), & n_U &= (a+1)(b+1), \\ E &= (d + rt + 1)(rs + 1). \end{aligned}$$

Define $\alpha = \|A\|_1$, $\beta = \|B\|_1$, $\delta = \|D\|_1 \geq 1$, and

$$H_P = \max_{0 \leq i \leq r} \left\{ \delta^{r-i} \prod_{j=0}^{i-1} (\alpha + 2jt\delta) \right\}, \tag{10}$$

$$H_U = \max\{1, \beta + 2(r-1)s\delta\}, \qquad H = \max\{H_P, H_U\}, \tag{11}$$

$$K = \min\{H^E, H_P^{n_P} H_U^{n_U}\}. \tag{12}$$

An empty product equals one. There exist $p_0, \ldots, p_r \in \mathbb{Z}[x]$, not all zero, and $U \in \mathbb{Z}[x,y]$, with

$$\deg_x p_i \leq d, \qquad \deg_x U \leq a, \qquad \deg_y U \leq b,$$

such that

$$\sum_{i=0}^{r} p_i(x) \partial_x^i f = \partial_y \left(\frac{U}{D^{r-1}} f \right). \tag{13}$$

Every integer coefficient of the p_i and U has absolute value at most K. In particular, the nonzero telescoper $P = \sum_{i=0}^{r} p_i(x)\partial_x^i$ has order at most r, degree at most d, and coefficient magnitude bit length at most

$$1 + \lceil \min\{E \log_2 H, n_P \log_2 H_P + n_U \log_2 H_U\} \rceil. \tag{14}$$

Its dense representation has at most n_P coefficients.

Proof. Define $N_0 = 1$ and, for $i \geq 0$,

$$N_{i+1} = D\partial_x N_i + (A - i\partial_x D) N_i. \tag{15}$$

Induction using (8) gives $\partial_x^i f / f = N_i / D^i$ and $\deg_x N_i \leq it$, $\deg_y N_i \leq is$. Thus $M_i = N_i D^{r-i}$ has bidegree at most (rt, rs).

Make a polynomial ansatz for the p_i and U with the bounds in the statement. After multiplication by D^r/f, identity (13) becomes

$$\sum_{i=0}^{r} p_i M_i - D\partial_y U - \big(B - (r-1)\partial_y D\big) U = 0. \tag{16}$$

Its bidegree is at most $(d+rt, rs)$. Equating coefficients gives a homogeneous integer matrix M with E rows and $n_P + n_U$ columns. Zero rows can be included.

We must ensure that a kernel vector has a nonzero p-component. The certificate-only equation is

$$D\partial_y U + (B - (r-1)\partial_y D)U = 0. \tag{17}$$

Over $\mathbb{Q}(x)$ its polynomial solution space has dimension at most one: the ratio of two nonzero solutions has zero y-derivative and therefore belongs to $\mathbb{Q}(x)$. If a bounded nonzero solution U_0 exists, choose one of its nonzero y-coefficients. Extracting that same coefficient from any bounded solution is injective, because every solution is $q(x)U_0$. Its image lies in $\mathbb{Q}[x]_{\le a}$. Hence the $\mathbb{Q}$-dimension of the bounded certificate-only kernel is at most $a+1$. Meanwhile

$$\begin{aligned}\dim_{\mathbb{Q}} \ker M - (a+1) &\ge n_P + n_U - E - (a+1)\\ &= (r-s)(d+1) - t\big((2r-1)s + r\big) > 0.\end{aligned}$$

Thus some kernel vector has a nonzero p-component. This argument includes zero logarithmic derivatives and the cases $s = 0$ or $t = 0$.

Next we bound column norms. Since $\|\partial_x N_i\|_1 \le it\|N_i\|_1$ and $\|\partial_x D\|_1 \le t\delta$, recurrence (15) yields

$$\|N_{i+1}\|_1 \le (\alpha + 2it\delta)\|N_i\|_1.$$

Consequently $\|M_i\|_1 \le H_P$. Each column belonging to a coefficient of p_i is a monomial translate of M_i, so its coefficient-vector ℓ_1 norm is at most H_P. For a coefficient of $x^j y^k$ in U, where $0 \le k \le b$, the corresponding column is, up to sign and an x-monomial translate, the coefficient vector of

$$kDy^{k-1} + (B - (r-1)\partial_y D)y^k.$$

Its norm is at most $k\delta + \beta + (r-1)s\delta \le H_U$.

Let $\rho = \operatorname{rank} M$. If $\rho = 0$, a unit vector in any p-coordinate proves the assertion. Otherwise choose ρ independent columns and ρ rows whose intersection C is nonsingular. For each nonpivot column k, give its free coordinate the value $\det C$, set the other free coordinates to zero, and set the pivot coordinates equal to $-\operatorname{adj}(C)M[I,k]$. These integer vectors solve all rows: the pivot columns span the full column space, and restriction to the selected rows is injective on that space. They form a basis of the rational kernel.

Every coordinate of such a vector is zero or a ρ-by-ρ minor of M. Hadamard's inequality, using that a column's ℓ_2 norm is at most its ℓ_1 norm, bounds each minor by $H^\rho \le H^E$. Since a minor uses distinct columns, it is also bounded by $H_P^{n_P} H_U^{n_U}$; both bases are at least one. At least one basis vector has a nonzero p-component, because the full kernel does. That vector proves (13) and (12); the bit estimate follows. □

Proposition 3.2. *Admissible choices in Theorem 3.1 are*

$$(r,d) = \big(s+1, t(2s^2+2s+1)\big) \quad \text{and} \quad (r,d) = \big(2s+1, t(4s+1)\big).$$

More generally, for any integer $r > s$ one may take

$$d = \left\lfloor \frac{t((2r-1)s + r)}{r-s} \right\rfloor.$$

If $t, s \ge 1$ and $h = \log_2 \max\{1, \|A\|_1, \|B\|_1, \|D\|_1\}$, the second uniform choice yields a telescoper with coefficient bit length

$$O\big(ts^3(h + \log(ts+2))\big).$$

Proof. For the first choice the difference between the two sides of the strict inequality in (9) equals one. For the second choice it equals $2ts+s+1$. For the floor choice, $d+1$ is strictly greater than the displayed quotient. For the asymptotic bound, $r = O(s)$, $d = O(ts)$, $n_P = O(ts^2)$, and $n_U = O(ts^3)$. Formula (10) gives $\log_2 H_P = O(s(h+\log(ts+2)))$, while $\log_2 H_U = O(h+\log(s+2))$. Substitute these bounds into the second term of (14). □

Remark 3.3. The assertion concerns existence of a bounded, possibly nonminimal-order telescoper. It neither bounds every scalar multiple nor promises minimum order. Common integer content can safely be divided out of P, with a rational rescaling of the certificate; no height claim is made for division by an arbitrary common polynomial factor. Rational exponents are included when the logarithmic derivatives lie in $\mathbb{Q}(x, y)$. An arithmetic bound for arbitrary irrational constants needs an explicit encoding and is not part of this statement. If the input polynomial coefficients have magnitude bit length at most c, then $h \le c + \log_2((t+1)(s+1))$.

Exact checks. The polynomial matrices in the proof were constructed over $\mathbb{Z}$ for seven examples. Exact nullspaces were computed with SymPy 1.14.0; the resulting polynomial identity, the certificate-only kernel bound, every column-norm estimate, and the final integer-height bound were checked exactly. Table 1 lists the actual operators selected by the computation, not claimed minimal operators. Here κ denotes the dimension of the bounded certificate-only kernel. All examples use the first uniform choice in Proposition 3.2; the third uses a deliberately nonsharp y-degree bound $s = 2$ to include a nonzero certificate-only kernel.

f	r	d	κ	Computed P
1	1	0	1	∂_x
e^{x^2}	1	1	2	$\partial_x - 2x$
$(x+y)^{-1}$	3	13	13	∂_x^3
$x+y$	2	5	7	∂_x^2
$e^{xy}/(y^2+1)$	4	25	0	$x(\partial_x^2+1)$
$e^{x/(x+y)}$	3	26	0	$x\partial_x^3 + \partial_x^2$
$(x+y)^{2/3}$	2	5	0	∂_x^2

Table 1: Exact validation of the polynomial ansatz and arithmetic bounds.

For example, with $f = e^{xy}/(y^2+1)$ we used $A = y^3 + y$, $B = x(y^2+1) - 2y$, $D = y^2 + 1$. The selected solution has $U = (y^2+1)^4$, so its certificate is $(y^2+1)f = e^{xy}$, and the identity is $x(\partial_x^2+1)f = \partial_y(e^{xy})$. For the moving-pole example the certificate numerator is $U = y^2(2x+y)$ with $D = (x+y)^2$ and $r = 3$. These checks supplement the proof; they do not establish novelty.

Interpretation. The theorem supplies an explicit arithmetic bound for all inputs in model (8), addressing the literal bound request of Problem 4 in that model. The bound is deliberately coarse and no matching lower bound is asserted. Its independent derivation and verification do not prove absence of an identical or stronger result in the literature. In particular, the stronger requirements of a historically new principle and an established first resolution remain unverified.

4 Reproducible checks and remaining limitations

The mathematical arguments above apply to all permitted inputs and indices. The following finite checks are supplementary regression checks for the constructions and for common implementation errors.

Two independently written coefficient-checking programs verified 465 exact coefficient comparisons in 16 rational constructions. The cases include overlapping and weighted index maps, zero rows, zero columns, an all-zero map, signed numerator coefficients, and vanishing constant terms. Each program constructs and cancels the rational divided differences before independently extracting coefficients. The denominator is checked to be nonzero at the origin.

The arithmetic-bound checker constructs seven exact integer matrices, including a 390×420 example and a 231×263 example. It checks an actual telescoper-certificate identity, the dimension of the certificate-only kernel, all column-norm bounds, and the stated integer coefficient bound. It also tests vanishing logarithmic derivatives, rational input functions, a rational exponent, and a moving pole.

Single-source reproduction. This TeX file contains three complete Python programs in `filecontents*` blocks. Compiling it with a current pdfLaTeX installation writes the programs alongside the PDF. With Python 3 and SymPy installed, execute

```
python verify_grouped_diagonals.py
python verify_grouped_diagonal.py
python validate_problem4.py
```

The first and third programs write JSON reports. No floating-point tolerance is used to accept a mathematical identity. The manuscript contains its own bibliography and needs no external TeX inputs or figures.

What is not established. The results do not prove that either construction is the first in the literature. They do not introduce a historically new basic principle. The arithmetic theorem does not assert an optimal bound or minimal-order coefficient bound, and it does not assign a bit size to unencoded complex constants. No further problem from the eighteen-item list is asserted to be resolved by this note.

References

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